

Module 5: Mobile Forensics

- What is SIM cards Forensic? Explain the SIM architecture and file structure? (05 Marks) 1
- Explain data analysis in mobile forensics? Also explain what type of evidence will be obtain from any social networking application (e.g Facebook, whatsapp, wechat) (10 Marks) 1
- Describe GPS evidentiary data. What are its challenges and limitations? How to extract waypoints and track points from GPX files for displaying the tracks on a Map? (10 Marks) 1
- Discuss the significance of ICCID, IMSI, MSISDN and LAI. (05 Marks) 2
- What is GPS forensic? Explain the structure of the GPS device. Explain GPS Exchange Format (GPX). (10 Marks) 2
- Explain what SIM cards Forensics means. Explain the SIM architecture and file structure. Explain evidence extraction in SIM card forensics. (10 Marks) 2
- Explain the SIM architecture and file structure? (05 Marks) 3
- What is Android forensic? Explain the structure of smartphone? (10 Marks) 3
- What is GPS forensic? Explain structure of GPS device? Explain GPS Exchange Format (GPX)? (10 Marks) 3

- **Explain the SIM architecture and file structure?**
- **What is Android forensics? Explain the structure of smartphone?**
- **What is GPS forensic? Explain structure of GPS device? Explain GPS Exchange Format (GPX)?**

D V T A C A D E M Y

Analysis Step in Digital Forensic Examination Process

Digital Forensics analysis is the stage where the collected digital evidence is examined carefully to identify useful information related to the investigation.

During this step, investigators:

- study files, logs, emails, and system records,
- recover deleted or hidden data,
- identify suspicious activities,
- and determine how the incident occurred.

The main objective of analysis is to reconstruct events, find the source of the attack or crime, and extract relevant evidence that can support the investigation or legal proceedings.

Significance ICCID, IMSI, MSISDN, LAI

09 May 2026 16:42

Significance of ICCID, IMSI, MSISDN, and LAI

In SIM Card Forensics and mobile communication systems, identifiers such as ICCID, IMSI, MSISDN, and LAI are important for identifying subscribers, SIM cards, and network locations. These identifiers help investigators analyze mobile devices and communication activities during forensic investigations.

1. ICCID (Integrated Circuit Card Identifier)

Integrated Circuit Card Identifier is a unique serial number assigned to every SIM card.

The ICCID identifies the physical SIM card and is usually stored on the SIM card itself. It helps mobile network operators track and manage SIM cards within their networks.

In forensic investigations, ICCID helps investigators identify the exact SIM card used in a mobile device and link it to a subscriber or communication activity.

2. IMSI (International Mobile Subscriber Identity)

International Mobile Subscriber Identity is a unique number used to identify a mobile subscriber in a cellular network.

The IMSI is stored inside the SIM card and contains:

- country code,
- network code,
- and subscriber identification number.

Mobile networks use the IMSI to authenticate subscribers and provide communication services.

In digital forensic investigations, IMSI helps investigators identify the subscriber associated with a mobile device and trace network usage activities.

3. MSISDN (Mobile Station International Subscriber Directory Number)

Mobile Station International Subscriber Directory Number refers to the actual mobile phone number assigned to a subscriber.

It is the number used for making calls and sending messages. The MSISDN helps identify and contact a mobile subscriber within the communication network.

During forensic investigations, MSISDN helps investigators link communication records, call logs, and message activities to a particular user or device.



4. LAI (Location Area Identity)

Location Area Identity is an identifier used in mobile networks to represent a specific geographical area covered by a group of cell towers.

The LAI helps the network determine the approximate location of a mobile device and manage subscriber mobility between different network areas.

In forensic analysis, LAI information helps investigators determine the geographical region where a mobile device was active during communication activities.

GPS evidentiary data

09 May 2026 16:41

GPS Evidentiary Data

GPS Forensics evidentiary data refers to the location-related information collected from GPS-enabled devices that can be used as digital evidence in investigations and legal proceedings. This data helps investigators determine the movement, location history, routes traveled, speed, direction, and timestamps associated with a person, vehicle, or object.

GPS evidentiary data is commonly obtained from:

- vehicle navigation systems,
- smartphones,
- portable GPS devices,
- fitness trackers,
- and fleet management systems.

The data collected from these devices can help investigators reconstruct events and verify timelines related to criminal or suspicious activities.

Types of GPS Evidentiary Data

1. Waypoints

Waypoints are user-defined geographical locations stored in the GPS device. They represent important locations such as destinations, checkpoints, or marked positions.

2. Track Points

Track points are sequential location coordinates automatically recorded by the GPS device during movement. They help investigators reconstruct the exact path traveled by a user or vehicle.

3. Routes

Routes represent planned navigation paths between multiple locations. They help identify intended travel directions and destinations.

4. Timestamp Information

GPS devices store date and time information along with location coordinates. This information is useful for creating timelines during investigations.

Challenges and Limitations of GPS Evidentiary Data

Although GPS evidence is valuable in investigations, several challenges and limitations affect its reliability and analysis.

1. Signal Interference and Loss

GPS signals may be affected by tall buildings, tunnels, weather conditions, or dense forests. Signal interruption can reduce location accuracy and create incomplete records.

2. Data Modification or Deletion

GPS data can be intentionally deleted, altered, or manipulated by users or attackers, making forensic analysis more difficult.

3. Limited Storage Capacity

Some GPS devices store only a limited amount of historical data. Older records may be automatically overwritten after storage becomes full.

4. Device Dependency

Different GPS devices use different file formats, storage methods, and software systems. This lack of standardization may complicate forensic investigations.

Extraction of Waypoints and Track Points from GPX Files for Displaying Tracks on a Map

GPS Exchange Format (GPX) files store GPS information such as waypoints, tracks, and routes. In Garmin nüvi devices, the current waypoints, tracks, and routes are stored in the unit's `Current.gpx` file. Garmin refers to waypoints as Favorites.

The following steps are used to extract waypoints and track points from GPX files and display them on a map:

1. Access the GPX File

The investigator first obtains the `Current.gpx` file from the GPS device. This file contains saved waypoints, routes, and track information that can be analyzed during the investigation.

2. Open the GPS Home Screen

To view the saved waypoints on the Garmin device:

- Go to the GPS home screen.
- Select the option “Where To?”

This option allows access to saved location information stored in the GPS device.

3. Select Favorites

In the next screen:

- Press on **"Favorites"**

Garmin devices store waypoints under the Favorites section.

4. Open All Favorites

After selecting Favorites:

- Press on **"All Favorites"**

This option displays all saved waypoint locations stored in the device.

5. Display the Waypoints

The GPS device then displays the saved favorites or waypoints along with their associated location information. These waypoints help investigators identify important visited locations.

Displaying the Tracks on a Map

Displaying GPS tracks on a map is important in forensic investigations because it helps demonstrate that the GPS device was present in a particular area on a specified date and time.

The extracted track points and route information from the GPX file can be displayed visually on digital maps using mapping applications and websites.



One commonly used application for displaying tracks is:

- Google Earth

The mapping software plots the extracted coordinates and connects the track points sequentially to display the complete travel route on the map. This visual representation helps investigators analyze movement patterns, travel history, and visited locations during the investigation.

SIM architecture and file structure *

09 May 2026 16:39

Simcard forensic Evidence extraction

SIM Card Forensics is a branch of digital forensics that involves the examination and analysis of SIM cards used in mobile devices to recover digital evidence.

It helps investigators collect information such as:

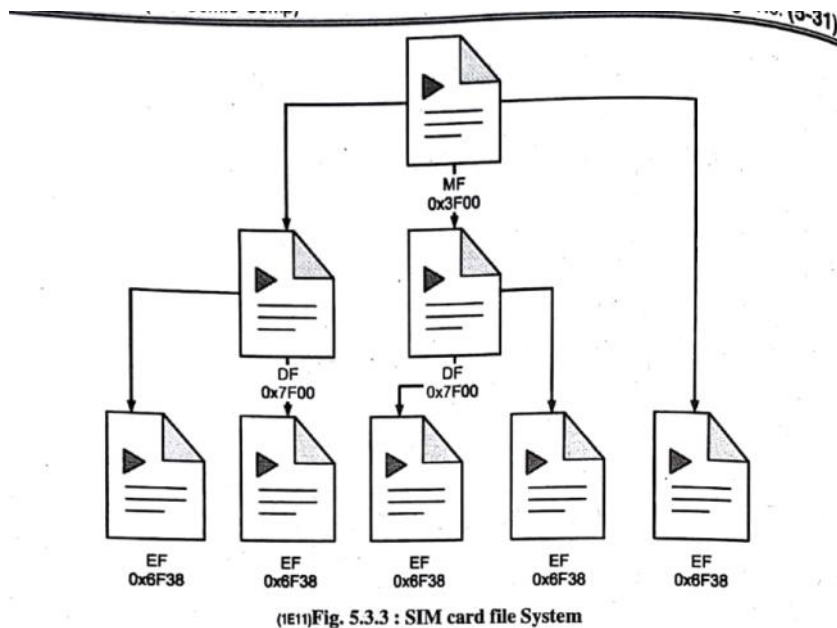
- contact numbers,
- SMS messages,
- call records,
- subscriber identity details,
- and network-related information stored on the SIM card.

SIM card forensics is commonly used in cyber crime investigations, fraud cases, and criminal investigations to trace communication activities and identify mobile users.

1. The **Subscriber Identity Module (SIM)** is a smart card used in mobile devices to store subscriber information and enable secure communication with the cellular network. It has a microprocessor, memory, and an operating system.
2. SIM architecture refers to how data is **organized, stored, and accessed** inside the SIM card using a **hierarchical file system** managed by the operating system inside the SIM.

1. Components of SIM

- **CPU (Microcontroller):**
Acts as the brain of the SIM card. It processes commands from the mobile device and controls all internal operations such as authentication, data access, and security functions.
- **ROM (Read Only Memory):**
Stores the SIM's operating system and essential programs. The data in ROM is permanent and cannot be modified.
- **RAM (Random Access Memory):**
Used for temporary storage during execution of tasks. It stores intermediate data while the SIM is performing operations.
- **EEPROM (Electrically Erasable Programmable ROM):**
A non-volatile memory used to store important user and network data such as IMSI, phonebook contacts, SMS, and authentication keys. Data is retained even when power is off.



2. File System Architecture

SIM uses a **hierarchical (tree-like) file system** to organize and manage data efficiently. It consists of three levels:

- **Master File (MF):**
The root directory of the SIM file system. It contains references to all other files and directories within the SIM.
- **Dedicated File (DF):**
These are subdirectories under the Master File. They are used to group related files together, such as telecom services or network-specific data.
- **Elementary File (EF):**
These are the actual data files where information is stored. Each EF contains specific data like subscriber identity, SMS, or contact details, and can be of different types (linear, cyclic, transparent).

5. Storage Formats of Elementary Files (EF)

Data within the Elementary Files can be stored in three distinct formats:

1. **Linear Fixed:** Data is stored in a sequence of records with a fixed length. Records are accessed by their record number.
2. **Cyclic:** A sequence of records where, after the last record is reached, the system cycles back to the first record (e.g., used for "recent calls").
3. **Transparent (Binary):** The most common format. It is a simple string of bytes with no fixed structure, allowing applications to save data in their own custom formats.

5.4.4 Evidence Extraction: Summary

Extracting data from the **EF file system** on a SIM card is critical for forensic investigations because the information often differs from what is stored on the handset itself.

Key Evidence Categories

- **Contacts (Abbreviated Dialing Numbers - ADN):**
 - SIMs can store between **250 to 600+ contacts** depending on memory size (32K to 128K).
 - These contacts are saved directly by the user and are not accessible to network providers, making them vital for identifying a suspect's relationships.
- **Call Logs:**
 - While phones log all call types, SIM cards primarily store **outgoing calls** (Last Dialed Numbers).
 - Investigators must extract these because the SIM logs may contain data different from the device's internal records.
- **SMS Messages:**
 - SIMs store text messages (up to 160 characters) using **GSM 03.38** or **Unicode** encoding.
 - **Note:** Deleted messages may still be recoverable from SIM memory, though messages from third-party apps (like Google Hangouts) are not stored there.

Android Forensics

Android Forensics is a branch of digital forensics that involves the identification, collection, preservation, examination, and analysis of data from Android devices such as smartphones and tablets for investigation purposes.

Android forensics is used to recover digital evidence related to calls, messages, emails, images, videos, application data, browsing history, location information, and social media activities stored on Android devices.

The main objective of Android forensics is to identify and analyze evidence that may help in cyber crime investigations, fraud detection, legal proceedings, and security incident analysis.

The forensic process generally includes:

- acquiring data from the device,
- preserving evidence integrity,
- analyzing stored and deleted data,
- and preparing forensic reports.

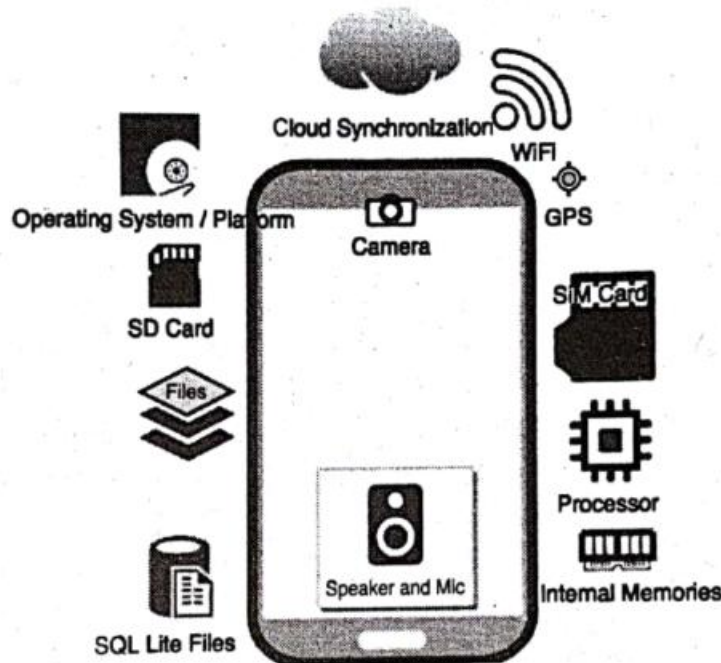
Android forensics faces challenges such as device encryption, screen locks, frequent operating system updates, and anti-forensic techniques used to hide or destroy evidence.

Thus, Android forensics plays an important role in modern digital investigations because mobile devices store a large amount of personal and communication data that can serve as valuable digital evidence.

Structure of smartphones *

09 May 2026 16:40

- The number of applications that are accessible for Mobile Phone Devices (e.g. smartphones) is growing rapidly, transforming our lives. Smartphones are becoming more powerful (e.g., rising CPU processing power, a wide variety of sensors built into it and user-friendly touch screen interface).
- A typical smartphone today has the logical structure and is made up of the following, as shown in Fig. 5.1.1.



(1E1)Fig. 5.1.1 : Basic Hardware Structure of Mobile Phones

- **Processor** : It is regarded as the smartphone's brain. Apple's A8, Qualcomm's Snapdragon 810, and Samsung's Exynos lineup are a few examples of mobile processors.
- **Memory Storage** : Modern smartphones offer a variety of memory storage options. For instance, the Android-powered Samsung Galaxy S7 allows both internal memory and external storage in the form of **microSD cards**. In addition, the internal memory of smartphones is frequently divided into two sections: System storage and Phone storage.
- **Sensors** : Unlike conventional computers, an increasing number of smartphones come with a range of potent sensors, including an accelerometer, digital compass, gravity sensor, gyroscope, GPS, fingerprint reader, and thermometer. These sensors produce verity of information and artefacts that may be vulnerable sometime.

- **Network Connectivity :** The number of networked apps is rapidly increasing today as a result of Wi-Fi being widely accessible on smartphones. For instance, the majority of smartphones come with **pre-installed social networking apps** like Facebook and Twitter. Additionally, cloud computing is showing up more and more on smartphones. Applications and data storage are offered to mobile users as services over the Internet in cloud computing.
- **Speakers and cameras :** Smartphones of the present day include speakers and cameras with very good precision and sensor integrations.
- However, there are currently innumerable variations of smartphones available. Although their internal designs are essentially very similar, phone manufacturers may store data in proprietary formats for their mobile phones, which makes it extremely difficult to forensically test mobile phones. However, the number of mobile operating systems is decreasing. Android is the most widely used.

GPS *

09 May 2026 16:40

1. GPS forensic
2. Structure of GPS device
3. GPX

GPS forensic

GPS Forensics

GPS Forensics is a branch of digital forensics that involves the identification, collection, examination, and analysis of location-related data obtained from Global Positioning System (GPS) devices.

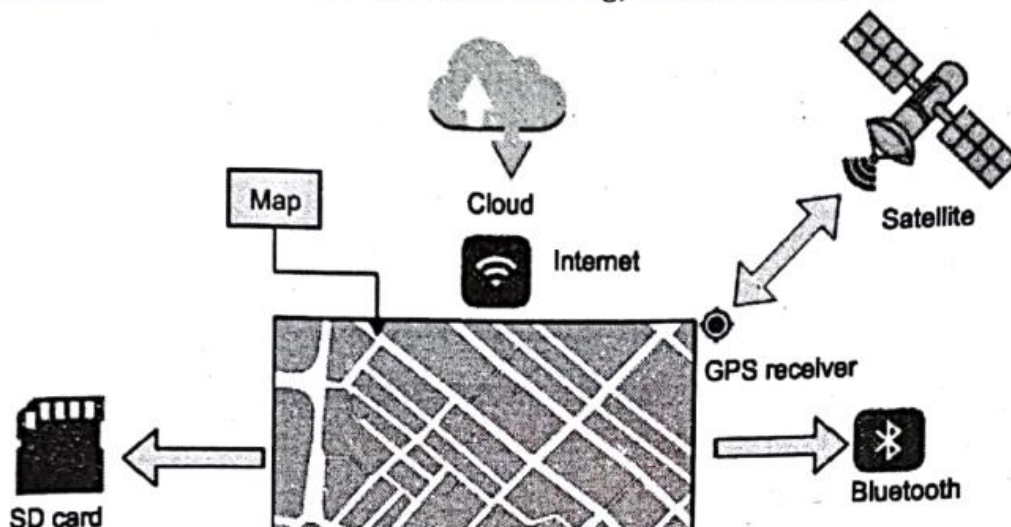
GPS forensics is used to determine the movement, location history, routes traveled, speed, and time information associated with a person or vehicle. It plays an important role in criminal investigations, accident analysis, missing person cases, and cyber crime investigations.

GPS evidence may be collected from:

- vehicle navigation systems,
- smartphones,
- fitness trackers,
- portable GPS devices,
- and fleet management systems.

Structure of device

- GPS device forensics can produce significant evidence in both civil and criminal cases. Personal GPS units, as well as devices for cars, planes, and ships, are examples of our contemporary GPS equipment. Additionally, GPS applications like Google Maps are increasingly common in modern smartphones. A common GPS device today has the logical structure and is made up of the following, as shown in Fig. 5.2.2:



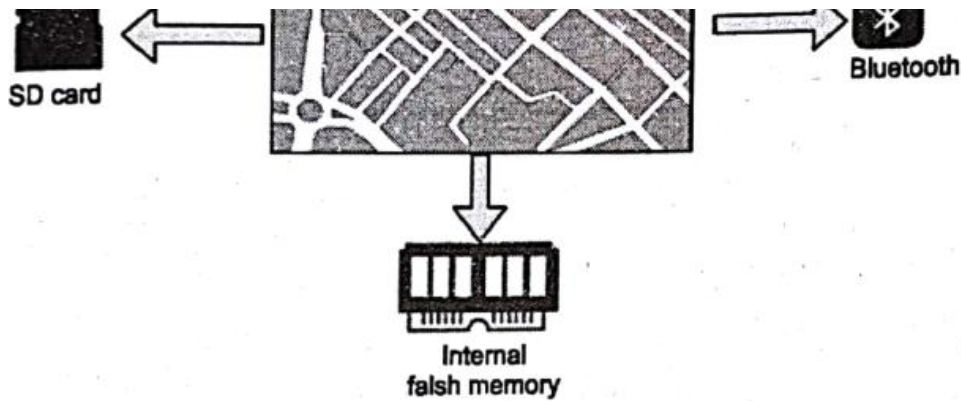


Fig. 5.2.2 : GPS device structure

- **GPS receiver** : The GPS receiver is an electrical device that uses radio waves transmitted by GPS satellites to calculate the user's current location.
- **Built-in map** : It gives the user access to a map view by mapping the position determined using the signals transmitted by the satellites to a real-world location. The user's speed can also be calculated using the device's motion. Additionally, it can infer how drivers behave behind the wheel.
- **Network compatibility** : GPS systems today come with a variety of wireless protocols. For instance, Bluetooth is becoming common in modern smartphones. It enables pairing of GPS gadgets with the user's phone. As a result, a number of cloud services, such as automatic map updating, can be used to connect GPS devices to the Internet. It is, however, not required to have it.

GPS Exchange Format (GPX)

GPS Exchange Format, commonly known as GPX, is a standard XML-based file format used to store and exchange GPS information between different devices and software applications.

The GPX format allows location data to be transferred and shared in a consistent and organized manner across multiple systems.

Information Stored in GPX

A GPX file stores several types of GPS-related information that are useful in forensic investigations.

1. Waypoints

Waypoints represent specific geographical locations marked by the user for navigation or reference purposes.

2. Routes

Routes define a planned navigation path between multiple locations and guide users from one point to another.

3. Tracks

Tracks record the actual path traveled by a person or vehicle along with location coordinates and timestamps. This information helps investigators reconstruct movement history accurately.

4. Time and Location Information

GPX files also contain details such as latitude, longitude, elevation, timestamps, speed, and movement patterns that are useful for timeline analysis.

Importance of GPX in GPS Forensics

GPX files are highly valuable in forensic investigations because they help investigators reconstruct travel routes, identify visited locations, analyze movement behavior, and verify timelines associated with criminal or suspicious activities.

The information stored in GPX files may serve as important digital evidence during legal investigations and court proceedings.

